

Research Protocol: Investigating causal relationship between 4th down attempt and winning in the National Football League

Historically, sports have been one of the few industries that emphasize intangible factors such as confidence, eye test and gut feeling over tangible evidence like statistics and trends. While this has traditionally been the case, the last few decades have seen a concerted effort by various sports franchises to incorporate data analytics into their decision-making processes. Growing up a fan of sports, one of the most obvious cases of this was when Bill Belichick, arguably the greatest coach in American team sports, decided to go for it (run a play rather than punt/kick a field goal) on 4th down and 2 in his own territory in the final minutes of a prime-time game. Though the attempt was unsuccessful and arguably led to their defeat, he admitted to utilizing analytics to inform him on a decision that football traditionalists would have thought to be wrong. This famous sports moment, referred to as the Belichick 4th and-2, has been credited as one of the sparks behind the increased frequency of 4th down attempts, driven by analytical reasoning. More coaches have since adopted this aggressive but data driven mentality. Thus, this also sparked my curiosity to investigate a potential relationship between 4th down decisions and winning.

Looking at the team with the most NFL 4th down attempts from the 2021 (when the 17-game schedule was implemented) to 2024 regular season, the Detroit Lions stand out. They have averaged 9.75 wins per season during that time span, which has been good enough to make the playoffs twice in four years, a significant improvement considering they had only made the playoffs three times in the previous 20 years. While there are obvious other factors that come into play, such as personnel changes, most people credit their aggressive coaching style, particularly going for it on 4th down, as a key factor in their cultural and competitive transformation under head coach Dan Campbell, who took over in 2021. Moreover, looking at the 2024 regular season, the top five teams in 4th down conversion percentage included four of the teams that reached their respective conference championship games, while the fifth was also a playoff team. This pattern supports the idea that there is a relationship between better 4th down decision making and winning in the NFL.

After carefully reviewing the literature on this topic there seems to be agreement around this general sentiment. Yam and Lopez used a propensity score model to match cohorts of teams that went for it on 4th down with those that did not, using a nearest-neighbor matching algorithm [4]. They then measured the Average Treatment Effect on the Control (ATE) and found that, among control teams (those that did not go for it on 4th down), a better 4th down strategy, which, in this context, refers to an increased likelihood of going for it, was worth roughly an additional 0.4 wins per year. They also found that the majority of NFL teams would have improved their records by attempting more 4th down conversions, averaging an increase of roughly 4.7 wins over a 13-season span (2004–2016). Romer, who approached the problem of 4th down conservatism from an economics perspective, used dynamic programming analysis on all regular-season NFL games from 1998 to 2000 to show that this conservatism is costing teams wins [3]. His analysis indicated that, out of 1,068 4th down situations where teams would have been better running a play, they instead chose to punt 959 times. He concluded that “teams' 4th down decisions systematically

and significantly deviated from choices that would have maximized their chances of winning”. Of course, these papers effectively examine improved 4th down strategy, rather than advocating for always running a play on 4th down. However, the overall sentiment in the literature still remains that in certain contexts, being more aggressive on 4th down leads to winning more games.

I plan to investigate the causal relationship between 4th down play calling and winning football games in the NFL through an observational study of 4th down play calls during the 2024 season. I am focusing only on the 2024 season because teams, players, and personnel change significantly from year to year. Additionally, I am only examining the regular season, as all teams are eligible to participate in those games. Furthermore, I will only analyze 4th down plays when the team had possession at or beyond midfield, as this helps mitigate the factor of fear of being scored on if a conversion attempt fails, which would be difficult to measure.

To investigate this causal relationship, I will evaluate the Average Treatment Effect for the Treated (ATT), the Average Treatment Effect for the Control (ATC), and the Average Treatment Effect (ATE), all of which represent a difference in risk, given that my outcome is binary (winning or not). For ATT and ATC, I will use matching methods with a propensity score model, while for ATE, I will apply Inverse Probability of Treatment Weighting (IPTW) using the same propensity score model.

My propensity score model will include the following variables:

- Exposure of interest: Running a play on 4th down
- Outcome: Winning the game
- Potential confounders: Home team, Score differential, Expected points (EP), Win probability added (WPA), Temperature, Wind, Stadium type, Offensive rating, Defensive rating of opposing team, Yard line, Goal-to-go distance
- Variables affecting only the outcome (to reduce type I error): Win probability (WP)
- Potential Results: Winning a game if you ran a play on 4th down vs punting/kicking a field goal on 4th down (Y^1 and Y^0 respectively)

Most of these variables were chosen based on their inclusion in previous similar studies [1], [2], [3] and [4]. The decision to incorporate additional variables was informed by football literature and an understanding of factors that can be reasonably measured and are believed to influence both the outcome and the treatment.

Since this is an observational study, I have aimed to adhere to the four causal presuppositions necessary to relate my statistical parameters to causal parameters, ensuring that my statistical associations can be interpreted as causal effects. However, a potential limitation of my study is that it may not fully satisfy the assumption of non-interference, as the exposure of one individual (a team attempting a 4th down play) could influence the outcome (winning the game) for another individual (the opposing team), since multiple 4th down plays may come from the same game. I will further investigate empirical positivity during the data-cleaning process, which will help guide decisions about which covariates to retain in my analysis. Finally, by using a propensity score model, I believe I can address concerns regarding conditional exchangeability.

For my analysis, I will primarily use the R package nflfastR to obtain play-by-play data. For any additional data not included in this package, I will supplement it with information from ESPN.com.

Firstly, I plan to potentially present a correlation matrix or a scatterplot between key variables such as winning percentage, 4th down attempts, playoff success to try to formally establish an association between 4th down attempts and success/winning in the NFL. I also plan to present Table 1, comparing my variables of interest stratified by exposure to determine whether there is any imbalance between the two groups. This will also help me better assess empirical positivity. Additionally, I will include a graph of the exposed and non-exposed groups, both before and after matching, to illustrate the extent to which matching has balanced the groups based on the covariates of interest. Although I have not yet conducted matching with my cohorts, I will prioritize sample size when selecting matching settings and emphasize achieving better balance between the groups, which will influence my choice of caliper settings. This process will be applied for matching in ATC and ATT. Similarly, for IPTW, I will construct Table 1 post-weighting, reporting the mean and standard deviation in both groups with standardized differences. Finally, I will examine any extraneous weights and adjust them accordingly using truncation.

Ultimately, I believe this analysis is important because it will allow us to evaluate three key aspects:

1. The overall effect of increased 4th down attempts on winning (ATE)
2. The effect of increased 4th down attempts on teams that do not traditionally attempt many 4th downs (ATC)
3. The effect of increased 4th down attempts on teams that already adopt an aggressive 4th down strategy (ATT)

Some limitations of my research include the presence of numerous unmeasured factors that influence 4th down decisions and winning games. These factors remain unmeasured either due to a lack of available data or because they are inherently difficult to quantify. In particular, Moskowitz identifies psychological factors such as risk aversion and hot-hand bias as potential confounders affecting both 4th down decision-making and game outcomes [2], but they cannot be directly measured. Additionally, because I am restricting my analysis to a specific scenario, my findings will not be applicable to all 4th down situations. However, the results should still provide valuable insight into 4th down decision-making in the NFL. Many coaches in the NFL still solely rely on the aforementioned intangible factors when deciding crucial 4th down plays and hopefully my research could help make more informed decisions in those scenarios rather than purely relying on intuition.

Finally, within the annex, I will define certain football jargon especially relating to my variables of interest in order to better illustrate their importance to the study.

References:

1. Blinkoff, J., Voeller, M., Graham, S., & Wilson, J. R. (Year). *Predictive modeling of 4th down conversion in Power 5 conferences: Football data analytics*. The Sport Journal. Retrieved from <https://thesportjournal.org/article/predictive-modeling-of-4th-down-conversion-in-power-5-conferences-football-data-analytics>
2. Moskowitz, T. and Wertheim, L. J. (2012), *Scorecasting: The Hidden Influences Behind How Sports Are Played and Games Are Won*, New York, NY, USA: Three Rivers Press.
3. Romer, D. (2006). *Do firms maximize? Evidence from professional football*. **Journal of Political Economy**, **114**(2), 340-365. <https://doi.org/10.1086/500238>
4. Yam, D., & Lopez, M. (2018). *Quantifying the causal effects of conservative fourth down decision making in the National Football League*. SSRN. <https://doi.org/10.2139/ssrn.3114242>